Technical Appendix - Assessing the Impact of Monetary and Macroprudential Policies on Israel's Housing Market: A DSGE Model Approach

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1 Households

The utility function is separable in consumption C_t (excl. housing), housing H_t and labor N_t.

$$U(C_t, H_t, L_t) = z_t \Gamma_c U(C_t - \zeta C_{t-1}) + j_t U(H_t) - U(N_t),$$

where Γ_c is a scalar (to insure that in SS the marginal utility from consumption is independent on habit). ζ is a degree of habit in consumption, z_t,and j_t are shocks to consumption and housing preferences. The housing bundle contains both ownership ($h_{o,t}$) and rent ($h_{r,t}$), which are imperfect substitutes.

$$H(h_{o,t}, h_{r,t}) = \left[\gamma h_{o,t}^{(\varepsilon-1)/\varepsilon} + (1-\gamma) h_{r,t}^{(\varepsilon-1)/\varepsilon} \right]^{\varepsilon/(\varepsilon-1)},$$

where $\varepsilon > 1$ is a intratemporal elasticity of substitution between ownership and rent. $0 < \gamma < 1$ is a relative weight of ownership in the CES function.

1.1 Lenders

The lender j ,$j \in [0, 1]$, acts in two capacities: as a household and as an investor. As a household, he demands consumption, rent, and ownership, works, and values leisure. As an investor, he purchases houses for investment and rents them out to retailers, who are the final suppliers of rental services in the economy. The investor buys houses to gain capital benefits to increase his consumption and does not derive direct utility from the houses purchased for investment. Since all households are identical within a group of lenders, we omit the index j.

The utility function of lenders is

$$E_0 \sum \beta^i \left(\Gamma_c \log(C_t - \zeta^c C_{t-1}) + j_t \log \left(\left[\gamma h_{o,t}^{(\varepsilon-1)/\varepsilon} + (1-\gamma) h_{r,t}^{(\varepsilon-1)/\varepsilon} \right]^{\varepsilon/(\varepsilon-1)} \right) - \frac{1}{1+\varrho} N_t^{1+\varrho} \right)$$

where $\Gamma_c = \frac{1-\zeta^c}{1-\beta\zeta^c}$. Therefore:

$$E_0 \sum \beta^i \left(z_t \Gamma_c \log(C_t - \zeta^c C_{t-1}) + \frac{j_t}{1-\vartheta} \log \left[\gamma h_{o,t}^{1-\vartheta} + (1-\gamma) h_{r,t}^{1-\vartheta} \right] - \frac{1}{1+\varrho} N_t^{1+\varrho} \right)$$

where $\vartheta = \frac{1}{\varepsilon}$.

The budget constraint:

$$\begin{aligned} & C_t + q_t(h_{o,t} + h_{inv,t}) + r_t h_{r,t} + R_{t-1} B_{t-1} = \\ = \quad & B_t + q_t(h_{o,t-1} + h_{inv,t-1}) + p_t^m h_{inv,t} + w_t N_t + Div_t - T_t \end{aligned}$$

where $h_{inv,t}(j) = h_{inv,t}$ represents the amount of houses purchased for investment (all investors are equal). q_t is the real price of houses (in terms of the price P_t of the consumption bundle C_t, which excludes the consumption of rent services).

The investors rent out the purchased houses for investment $h_{inv,t}(j) = h_{inv,t}$ to retailers at the nominal price P_t^m at the beginning of period t as an intermediate good and receive the houses back at the end of period t. Therefore, the capital gains or losses related to reselling $h_{inv,t}$ in period $t+1$ fall solely only on the investors. $p_t^m(j) = p_t^m$ is a real price of rent received from the retailers (in terms of P_t, where $p_t^m = \frac{P_t^m}{P_t}$). Note that r_t is the real price of rent in the economy and $p_t^m \neq r_t$. The nominal price of rent is $P_t^{rent} = r_t P_t$.

$h_{o,t}$ is the amount of homes purchased for ownership, $h_{r,t}$ is the amount of houses for rent, B_t is the amount of one-period deposit, R_t is the gross real interest rate paid on one-period deposit, $R_t = r_t^{CB} - \pi_{t+1}$, r_t^{CB} is the nominal interest rate of the central bank and π_{t+1} is the expected inflation, w_t is the real wage, N_t is the number of hours worked. Div and T are dividends and taxes. We assume that there is no government, therefore $T_t = 0$.

The Lagrangian:

$$\begin{aligned} L_t \quad = \quad & E_0 \sum \beta^i \left(z_{t+i} \Gamma_c \log(C_{t+i} - \zeta^c C_{t+i-1}) + \frac{j_{t+i}}{1-\vartheta} \log \left[\gamma h_{o,t+i}^{1-\vartheta} + (1-\gamma) h_{r,t+i}^{1-\vartheta} \right] - \frac{1}{1+\varrho} N_{t+i}^{1+\varrho} \right) - \\ & -E_0 \sum \beta^i \lambda_{t+i} \left(\begin{array}{c} C_{t+i} + q_{t+i}(h_{o,t+i} + h_{inv,t+i}) + r_{t+i} h_{r,t+i} + R_{t+i-1} B_{t+i-1} - B_{t+i} \\ -q_{t+i}(h_{o,t+i-1} + h_{inv,t+i-1}) \\ -p_{t+i}^m h_{inv,t+i} - w_{t+i} N_{t+i} - Div_{t+i} + T_{t+i} \end{array} \right) \end{aligned}$$

Lenders have to decide about 6 variables: $C_t, h_{o,t}, h_{r,t}, B_t, N_t, h_{inv,t}$.

F.O.C of the lenders

(1) $C_t : \Gamma_c(\frac{z_t}{C_t - \zeta^c C_{t-1}} - \beta\zeta^c \frac{z_{t+1}}{C_{t+1} - \zeta^c C_t}) - \lambda_t = 0 =>$

$$\lambda_t = \Gamma_c\left(\frac{z_t}{C_t - \zeta^c C_{t-1}} - \beta\zeta^c \frac{z_{t+1}}{C_{t+1} - \zeta^c C_t}\right)$$

(2) $h_{o,t} : j_t \frac{\gamma h_{o,t}^{-\vartheta}}{\gamma h_{o,t}^{1-\vartheta} + (1-\gamma) h_{r,t}^{1-\vartheta}} - \lambda_t q_t + \beta \lambda_{t+1} q_{t+1} = 0 =>$

$$U_{h_o,t} = \lambda_t q_t - \beta \lambda_{t+1} q_{t+1}$$

where $U_{h_o,t} = j_t \frac{\gamma h_{o,t}^{-\vartheta}}{\gamma h_{o,t}^{1-\vartheta} + (1-\gamma) h_{r,t}^{1-\vartheta}}$ is the marginal utility from ownership.

(3): $h_{r,t} : j_t \frac{(1-\gamma) h_{r,t}^{-\vartheta}}{\gamma h_{o,t}^{1-\vartheta} + (1-\gamma) h_{r,t}^{1-\vartheta}} - \lambda_t r_t = 0 >$

$$U_{h_r,t} = \lambda_t r_t$$

where $U_{h_r,t} = j_t \frac{(1-\gamma) h_{r,t}^{-\vartheta}}{\gamma h_{o,t}^{1-\vartheta} + (1-\gamma) h_{r,t}^{1-\vartheta}}$ is the marginal utility from rent.

(4) $B_t : \lambda_t - \beta \lambda_{t+1} R_t = 0 >$

$$\lambda_t = \beta \lambda_{t+1} R_t$$

(5) $N_t : -\frac{1+\varrho}{1+\varrho} N_t^{\varrho} - \lambda_t(-w_t) = 0 =>$

$$N_t^{\varrho} = \lambda_t w_t$$

(6) $h_{inv,t} : -\lambda_t (q_t - p_t^m) + \beta \lambda_{t+1} q_{t+1} = 0 =>$

$$\lambda_t (q_t - p_t^m) = \beta \lambda_{t+1} q_{t+1}$$

Or

$$p_t^m = q_t - sdf_{t,t+1} q_{t+1}$$

where $sdf_{t,t+1} = \beta \frac{\lambda_{t+1}}{\lambda_t}$.

If we divide both sides by r_t

$$\frac{p_t^m}{r_t} = \frac{q_t}{r_t} - sdf_{t,t+1} \frac{q_{t+1}}{r_t}$$

Now, from the FOC of lenders in (2) and (3) we obtain

$$\frac{U_{h_o,t}}{U_{h_r,t}} = \frac{q_t}{r_t} - sdf_{t,t+1} \frac{q_{t+1}}{r_t}$$

Combining two last expressions we can link between p_t^m and r_t.

$$\frac{U_{h_o,t}}{U_{h_r,t}} = \frac{q_t}{r_t} - sdf_{t,t+1} \frac{q_{t+1}}{r_t} = \frac{p_t^m}{r_t}$$

Thus

$$p_t^m = \frac{U_{h_o,t}}{U_{h_r,t}} r_t$$

Finally can represent the pricing formula

$$q_t = E_t \sum_{i=0}^{\infty} sdf_{t,t+i} p_{t+1}^m$$

which is a discounted value of future payoffs from rent, where $sdf_{t,t} \equiv 1$, $sdf_{t,t+i} = \beta^i \frac{\lambda_{t+i}}{\lambda_t}$ is a stochastic discount factor of the lenders. Note that the ratio below $\frac{p_t^m}{q_t}$ is not equal to the rent-to-price ratio in the economy, $\frac{r_t}{q_t}$, because $p_t^m \neq r_t$.

The alternative representation is by using $p_t^m = \frac{U_{h_o,t}}{U_{h_r,t}} r_t$:

$$q_t = E_t \sum_{i=0}^{\infty} sdf_{t,t+i} \frac{U_{h_o,t+i}}{U_{h_r,t+i}} r_{t+i}$$

1.2 Retailers

Retailers purchase houses from investors at the price P_t^m, and these houses are used as intermediate goods. The final product of houses used for rent in the economy, Y_t^{rent}, is a composite of a continuum of mass unity of differentiated retail houses that use intermediate houses from investors as the sole input (i.e., $h_{inv,t}(j) = Y_{f,t}$, where $Y_{f,t}$ is the amount of houses of retailer f and $h_{inv,t}(j)$ is the intermediate product of investor j). The final product of houses Y_t^{rent} used for rent in the economy is

$$Y_t^{rent} = \left[\int_0^1 \left(Y_{f,t} \right)^{\frac{\eta_r - 1}{\eta_r}} df \right]^{\frac{\eta_r}{\eta_r - 1}}$$

where η_r is the elasticity of substitution between differentiated houses. The demand for a differentiated retail house f is given by:

$$Y_{f,t} = \left(\frac{P_{f,t}^{rent}}{P_t^{rent}} \right)^{-\eta_r} Y_t^{rent}$$

The nominal price of Y_t^{rent} is given by:

$$P_t^{rent} = \left[\int_0^1 \left(P_{f,t}^{rent} \right)^{1-\eta_r} df \right]^{\frac{1}{1-\eta_r}}$$

The differentiated retail firms face a price rigidity of Rotemberg (1982). The expected profits of retailer is:

$$\max E_t \sum_{i=0}^{\infty} sdf_{t,t+i}^r x_{f,t+i},$$

where $x_{f,t}$ is a profit of retailer f expressed in real units of rental services, and $sdf_{t,t+i}^r = sdf_{t,t+i} \frac{r_{t+i}}{r_t}$, a stochastic discount factor of the lenders (owners of the retail firms) in terms of the marginal utility of rental services. More specifically, $sdf_{t,t+i}^r = \beta^i \frac{U'_{r,t+i}}{U'_{r,t}}$ ($U'_{r,t}$ is the marginal utility of rental services) and then using $U_{h_r,t} = \lambda_t r_t$, we obtain $sdf^r_{t,t+i} = sdf_{t,t+i}\frac{r_{t+i}}{r_t}$.

$$x_{f,t} = \frac{P^{rent}_{f,t}}{P^{rent}_t}Y_{f,t} - \frac{P^m_t}{P^{rent}_t}Y_{f,t} - \frac{\Theta}{2}\left(\frac{P^{rent}_{f,t}}{P^{rent}_{f,t-1}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_{t-1}}{P^{rent}_{t-2}}\right)^{\phi}\right)^2 Y^{rent}_t$$

where π is the SS rent inflation rate, ϕ is the degree of indexation to past rent inflation, and Θ is the cost associated with changing prices.

The expected profit is

$$\max E_t \sum_{i=0}^{\infty} sdf^r_{t,t+i}\left[\begin{array}{c} \frac{P^{rent}_{f,t+i}}{P^{rent}_{t+i}}Y_{f,t+i} - \frac{P^m_{t+i}}{P^{rent}_{t+i}}Y_{f,t+i} - \\ \frac{\Theta}{2}\left(\frac{P^{rent}_{f,t+i}}{P^{rent}_{f,t+i-1}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_{t+i-1}}{P^{rent}_{t+i-2}}\right)^{\phi}\right)^2 Y^{rent}_{t+i}\end{array}\right]$$

After inserting demand for $Y_{f,t}$ we get

$$\max E_t \sum_{i=0}^{\infty} sdf^r_{t,t+i}\left[\begin{array}{c} \left(\frac{P^{rent}_{f,t+i}}{P^{rent}_{t+i}}\right)^{1-\eta_r} Y^{rent}_{t+i} - \frac{P^m_{t+i}}{P^{rent}_{t+i}}\left(\frac{P^{rent}_{f,t+i}}{P^{rent}_{t+i}}\right)^{-\eta_r} Y^{rent}_{t+i} - \\ \frac{\Theta}{2}\left(\frac{P^{rent}_{f,t+i}}{P^{rent}_{f,t+i-1}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_{t+i-1}}{P^{rent}_{t+i-2}}\right)^{\phi}\right)^2 Y^{rent}_{t+i}\end{array}\right]$$

or

$$\max E_t \sum_{i=0}^{\infty} sdf^r_{t,t+i}\left[\begin{array}{c} \left(\frac{P^{rent}_{f,t+i}}{P^{rent}_{t+i}}\right)^{1-\eta_r} Y^{rent}_{t+i} - P^m_{t+i}\left(P^{rent}_{f,t+i}\right)^{-\eta_r}\left(\frac{1}{P^{rent}_{t+i}}\right)^{1-\eta_r} Y^{rent}_{t+i} - \\ -\frac{\Theta}{2}\left(\frac{P^{rent}_{f,t+i}}{P^{rent}_{f,t+i-1}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_{t+i-1}}{P^{rent}_{t+i-2}}\right)^{\phi}\right)^2 Y^{rent}_{t+i}\end{array}\right]$$

The first-order condition w.r.t. price $P^{rent}_{f,t}$

$$\begin{gathered}(1-\eta_r)\left(P^{rent}_{f,t}\right)^{-\eta_r}\left(\frac{1}{P^{rent}_t}\right)^{1-\eta_r} Y^{rent}_t + \eta_r P^m_t\left(P^{rent}_{f,t}\right)^{-\eta_r-1}\left(\frac{1}{P^{rent}_t}\right)^{1-\eta_r} Y^{rent}_t - \\ \Theta\left(\frac{P^{rent}_{f,t}}{P^{rent}_{f,t-1}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_{t-1}}{P^{rent}_{t-2}}\right)^{\phi}\right) Y^{rent}_t \frac{1}{P^{rent}_{f,t-1}} + sdf^r_{t,t+1}\Theta\left(\frac{P^{rent}_{f,t+1}}{P^{rent}_{f,t}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_t}{P^{rent}_{t-1}}\right)^{\phi}\right) Y^{rent}_{t+1}\frac{P^{rent}_{f,t+1}}{(P^{rent}_{f,t})^2} = 0\end{gathered}$$

Now let us multiply by $P^{rent}_{f,t}$

$$\begin{gathered}(1-\eta_r)\left(P^{rent}_{f,t}\right)^{1-\eta_r}\left(\frac{1}{P^{rent}_t}\right)^{1-\eta_r} Y^{rent}_t + \eta_r P^m_t\left(P^{rent}_{f,t}\right)^{-\eta_r}\left(\frac{1}{P^{rent}_t}\right)^{1-\eta_r} Y^{rent}_t - \\ \Theta\left(\frac{P^{rent}_{f,t}}{P^{rent}_{f,t-1}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_{t-1}}{P^{rent}_{t-2}}\right)^{\phi}\right) Y^{rent}_t \frac{P^{rent}_{f,t}}{P^{rent}_{f,t-1}} + sdf^r_{t,t+1}\Theta\left(\frac{P^{rent}_{f,t+1}}{P^{rent}_{f,t}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_t}{P^{rent}_{t-1}}\right)^{\phi}\right) Y^{rent}_{t+1}\frac{P^{rent}_{f,t+1}P^{rent}_{f,t}}{(P^{rent}_{f,t})^2} = 0\end{gathered}$$

After simplifications

$$\begin{gathered}(1-\eta_r)\left(\frac{P^{rent}_{f,t}}{P^{rent}_t}\right)^{1-\eta_r} + \eta_r\frac{P^m_t}{P^{rent}_t}\left(\frac{P^{rent}_{f,t}}{P^{rent}_t}\right)^{-\eta_r} - \\ -\Theta\frac{P^{rent}_{f,t}}{P^{rent}_{f,t-1}}\left(\frac{P^{rent}_{f,t}}{P^{rent}_{f,t-1}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_{t-1}}{P^{rent}_{t-2}}\right)^{\phi}\right) + sdf^r_{t,t+1}\frac{P^{rent}_{f,t+1}}{P^{rent}_{f,t}}\Theta\left(\frac{P^{rent}_{f,t+1}}{P^{rent}_{f,t}} - (\pi)^{1-\phi}\left(\frac{P^{rent}_t}{P^{rent}_{t-1}}\right)^{\phi}\right)\frac{Y^{rent}_{t+1}}{Y^{rent}_t} = 0\end{gathered}$$

Now let us define $\pi_t^{rent} = \frac{P_{f,t}^{rent}}{P_{f,t-1}^{rent}} = \frac{P_t^{rent}}{P_{t-1}^{rent}}$.Since $P_{f,t}^{rent} = P_t^{rent}$, (all firms are identical w.r.t. price setting)

$$1 - \eta_r + \eta_r \frac{P_t^m}{P_t^{rent}} -$$
$$-\Theta \pi_t^{rent} \left(\pi_t^{rent} - (\pi)^{1-\phi} \left(\pi_{t-1}^{rent} \right)^{\phi} \right) + sdf_{t,t+1}^r \pi_{t+1}^{rent} \Theta \left(\pi_{t+1}^{rent} - (\pi)^{1-\phi} \left(\pi_t^{rent} \right)^{\phi} \right) \frac{Y_{t+1}^{rent}}{Y_t^{rent}} = 0$$

After simplifications

$$\frac{1-\eta_r}{\eta_r} + \frac{P_t^m}{P_t^{rent}} -$$
$$-\frac{\Theta}{\eta_r} \pi_t^{rent} \left(\pi_t^{rent} - (\pi)^{1-\phi} \left(\pi_{t-1}^{rent} \right)^{\phi} \right) + sdf_{t,t+1}^r \pi_{t+1}^{rent} \frac{\Theta}{\eta_r} \left(\pi_{t+1}^{rent} - (\pi)^{1-\phi} \left(\pi_t^{rent} \right)^{\phi} \right) \frac{Y_{t+1}^{rent}}{Y_t^{rent}} = 0$$

$$\frac{P_t^m}{P_t^{rent}} = \frac{\eta_r - 1}{\eta_r} + \frac{\Theta}{\eta_r} \pi_t^{rent} \left(\pi_t^{rent} - (\pi)^{1-\phi} \left(\pi_{t-1}^{rent} \right)^{\phi} \right) - sdf_{t,t+1}^r \frac{\Theta}{\eta_r} \pi_{t+1}^{rent} \left(\pi_{t+1}^{rent} - (\pi)^{1-\phi} \left(\pi_t^{rent} \right)^{\phi} \right) \frac{Y_{t+1}^{rent}}{Y_t^{rent}}$$

In terms of already existing variables in the model we divide numerator and denominator of the LHS by P_t

$$\frac{P_t^m / P_t}{P_t^{rent} / P_t} = \frac{\eta_r - 1}{\eta_r} + \frac{\Theta}{\eta_r} \pi_t^{rent} \left(\pi_t^{rent} - (\pi)^{1-\phi} \left(\pi_{t-1}^{rent} \right)^{\phi} \right) - sdf_{t,t+1}^r \frac{\Theta}{\eta_r} \pi_{t+1}^{rent} \left(\pi_{t+1}^{rent} - (\pi)^{1-\phi} \left(\pi_t^{rent} \right)^{\phi} \right) \frac{Y_{t+1}^{rent}}{Y_t^{rent}}$$

After simplification

$$\frac{p_t^m}{r_t} = \frac{\eta_r - 1}{\eta_r} + \frac{\Theta}{\eta_r} \pi_t^{rent} \left(\pi_t^{rent} - (\pi)^{1-\phi} \left(\pi_{t-1}^{rent} \right)^{\phi} \right) - sdf_{t,t+1}^r \frac{\Theta}{\eta_r} \pi_{t+1}^{rent} \frac{Y_{t+1}^{rent}}{Y_t^{rent}} \left(\pi_{t+1}^{rent} - (\pi)^{1-\phi} \left(\pi_t^{rent} \right)^{\phi} \right)$$

where $\pi_t^{rent} = \frac{P_t^{rent}}{P_{t-1}^{rent}} = \frac{r_t P_t}{r_{t-1} P_{t-1}} = \frac{r_t}{r_{t-1}} \pi_t$.

Finally, we obtain the Phillips curve for rent prices:

$$\begin{aligned} \frac{p_t^m}{r_t} &= \frac{\eta_r - 1}{\eta_r} + \frac{\Theta}{\eta_r} \frac{r_t}{r_{t-1}} \pi_t \left(\frac{r_t}{r_{t-1}} \pi_t - (\pi)^{1-\phi} \left(\frac{r_{t-1}}{r_{t-2}} \pi_{t-1} \right)^{\phi} \right) \\ & \quad - sdf_{t,t+1}^r \frac{\Theta}{\eta_r} \frac{r_{t+1}}{r_t} \pi_{t+1} \frac{Y_{t+1}^{rent}}{Y_t^{rent}} \left(\frac{r_{t+1}}{r_t} \pi_{t+1} - (\pi)^{1-\phi} \left(\frac{r_t}{r_{t-1}} \pi_t \right)^{\phi} \right) \end{aligned}$$

Note that in the SS:

$$\frac{p^m}{r} = \frac{\eta_r - 1}{\eta_r} < 1$$

meaning that in SS, $r = \frac{\eta_r}{\eta_r - 1} p^m$, that is, the final price of rent on the economy is larger than the price of intermediate rent by markup.

Recall that the nominal profit of the retailers is

$$x_{f,t} = P_{f,t}^{rent} Y_{f,t} - P_t^m Y_{f,t} - \frac{\Theta}{2} \left(\frac{P_{f,t}^{rent}}{P_{f,t-1}^{rent}} - (\pi)^{1-\phi} \left(\frac{P_{t-1}^{rent}}{P_{t-2}^{rent}} \right)^{\phi} \right)^2 P_t^{rent} Y_t^{rent}$$

Dividends of the retailers in real terms of P_t are transferred to the lenders (investors)

$$Div_t^{retail} = r_t Y_t^{rent} - p_t^m Y_t^{rent} - \frac{\Theta}{2}\left(\pi_t^{rent} - (\pi)^{1-\phi}\left(\pi_{t-1}^{rent}\right)^{\phi}\right)^2 r_t Y_t^{rent},$$

where $Y_t^{rent} = h_{inv,t}$ (see proof below) and $\pi_t^{rent} = \frac{r_t}{r_{t-1}}\pi_t$.
In SS:

$$Div^{retail} = (r - p^m)Y^{rent} = (r - p^m)h_{inv}$$

What is the relationship between $Y_t^{rent}, h_{inv,t}$ and $Y_{f,t}$?

We defined previously the composite good of houses Y_t^{rent}, which is an aggregate of intermediate retailed houses where $h_{inv,t}(j) = Y_{f,t}$. Since $h_{inv,t}(j)$ is the same for all investors j, we have $\int_0^1 h_{inv,t}(j)dj = h_{inv,t}$. The main interest is to determine Y_t^{rent}. The supply of all intermediate goods of houses should be equal to the total demand for retailed houses, that is:

$$\int_0^1 h_{inv,t}(j)dj = h_{inv,t} = \int_0^1 Y_{f,t}df$$

Using the definition for $Y_{f,t}$ we have:

$$h_{inv,t} = \int_0^1 Y_{f,t}df = \int_0^1 \left(\frac{P_{f,t}^{rent}}{P_t^{rent}}\right)^{-\eta_r} Y_t^{rent}df = \int_0^1 \left(\frac{P_{f,t}^{rent}}{P_t^{rent}}\right)^{-\eta_r} dfY_t^{rent} = Y_t^{rent}$$

1.3 Borrowers

For simplicity, we assume that some parameters are equal between lenders and borrowers.: $\zeta^c = {\zeta'}^c, \vartheta = \vartheta', \varrho = \varrho'$.

The utility function is (denote with symbol ').

$$E_0 \sum \beta'^i \left(z_t'\Gamma_c' \log(C_t' - \zeta^c C_{t-1}') + \frac{j_t'}{1-\vartheta}\log\left[\gamma' {h'}_{o,t}^{1-\vartheta} + (1-\gamma'){h'}_{r,t}^{1-\vartheta}\right] - \frac{1}{1+\varrho}{N'}_t^{1+\varrho}\right)$$

where $\Gamma_c' = \frac{1-\zeta^c}{1-\beta'\zeta^c}$.
The budget constraint of the borrowers

$$C_t' + q_t h_{o,t}' + r_t h_{r,t}' + (R_{t-1}^L - 1 + \Upsilon)B_{t-1}' = B_t^{'NEW} + q_t h_{o,t-1}' + w_t' N_t'$$

where Υ denotes the constant amortization rate for determining the principal payments out of the stock of mortgage debt. So, the stock of debt in period t equals to the stock of debt in period $t-1$ after amortization. $B_t^{'NEW}$ is a new debt in period t.

$$B_t^{'} = (1-\Upsilon)B_{t-1}^{'} + B_t^{'NEW}$$

We assume that borrowers do not receive any dividends because they do not own any firms, so $Div_t' = 0$. We also assume there are no taxes because there is no government, so $T_t' = 0$.The (gross) interest rate on mortgages is R_t^L, where either (1) $R_t^L = R_t + \Psi$, or (2) $R_t^L = \frac{B_t^{'NEW}}{B_t^{'}}(R_t + \Psi) + (1 - \frac{B_t^{'NEW}}{B_t^{'}})R_{t-1}^L$,where Ψ is a constant spread over the riskless real interest rate R_t.

As in Alpanda and Zubairy (2017), borrowers are subject to a maximum Loan-to-Value LTV restriction on a new debt that we assume are always binding. The modified equation for credit supply is (see detailed explanation in the article):

$$B_t^{'NEW} = LTV_t q_t (h_{o,t}^{'} - h_{o,t-1}^{'}) + k\left(LTV_t q_t h_{o,t-1}^{'} - (1-\Upsilon)B_{t-1}^{'}\right)$$

or

$$B_t^{'NEW} = LTV_t q_t h_{o,t}^{'} - LTV_t q_t h_{o,t-1}^{'} + kLTV_t q_t h_{o,t-1}^{'} - k(1-\Upsilon)B_{t-1}^{'}$$

Finally

$$B_t^{'NEW} = LTV_t q_t h_{o,t}^{'} + (k-1)\, LTV_t q_t h_{o,t-1}^{'} - k(1-\Upsilon)B_{t-1}^{'}$$

where LTV_t follows exogenous process (ϵ_t^{LTV} is the policy choice).
We assume no depreciation in houses.
The stock of debt and new debt in the SS are given below:

$$\begin{aligned} B^{'} &= \left\{\frac{k}{\Upsilon + k(1-\Upsilon)}\right\} LTVqh_o^{'} \\ B^{'NEW} &= \Upsilon B' = \left\{\frac{\Upsilon k}{\Upsilon + k(1-\Upsilon)}\right\} LTVqh_o^{'} \end{aligned}$$

The Lagrangian:

$$\begin{aligned} L_t' &= E_0 \sum \beta'^i \left(z_{t+i}' \Gamma_c' \log(C_{t+i}' - \zeta^c C_{t+i-1}') + \frac{j_{t+i}'}{1-\vartheta} \log\left[\gamma' h_{o,t+i}'^{1-\vartheta} + (1-\gamma') h_{r,t+i}'^{1-\vartheta}\right] - \frac{1}{1+\varrho} N_{t+i}'^{1+\varrho} \right) - \\ &- E_0 \sum \beta'^i \lambda_{t+i}^{'} \left(\begin{array}{c} C_{t+i}' + q_{t+i} h_{o,t+i}' + r_{t+i} h_{r,t+i}' + (R_{t+i-1}^L - 1 + \Upsilon) B_{t+i-1}' \\ -B_{t+i}^{'NEW} - q_{t+i} h_{o,t+i-1}' - w_{t+i}' N_{t+i}' \end{array} \right) - \\ &- E_0 \sum \beta'^i \mu_{t+i}' \left(B_{t+i}^{'NEW} - LTV_{t+i} q_{t+i} h_{o,t+i}' - (k-1)\, LTV_{t+i} q_{t+i} h_{o,t+i-1}' + k(1-\Upsilon) B_{t+i-1}^{'} \right) \\ &- E_0 \sum \beta'^i \Lambda_{t+i}' \left(R_{t+i}^L - \frac{B_{t+i}^{'NEW}}{B_{t+i}^{'}} (R_{t+i} + \Psi) - (1 - \frac{B_{t+i}^{'NEW}}{B_{t+i}^{'}}) R_{t+i-1}^L \right) \end{aligned}$$

Now we substitute $B_{t+i}^{'NEW} = B_{t+i}^{'} - (1-\Upsilon)B_{t+i-1}^{'}$.

$$\begin{aligned}
L_t' \quad = \quad & E_0 \sum \beta'^i \left(z_{t+i}'\Gamma_c' \log(C_{t+i}' - \zeta^c C_{t+i-1}') + \frac{j_{t+i}'}{1-\vartheta} \log\left[\gamma' h_{o,t+i}'^{1-\vartheta} + (1-\gamma')h_{r,t+i}'^{1-\vartheta}\right] - \frac{1}{1+\varrho} N_{t+i}'^{1+\varrho} \right) - \\
& -E_0 \sum \beta'^i \lambda_{t+i}^{'} \left(\begin{array}{c} C_{t+i}' + q_{t+i}h_{o,t+i}' + r_{t+i}h_{r,t+i}' + (R_{t+i-1}^L - 1 + \Upsilon)B_{t+i-1}' \\ -B_{t+i}^{'} + (1-\Upsilon)B_{t+i-1}^{'} - q_{t+i}h_{o,t+i-1}' - w_{t+i}'N_{t+i}' \end{array} \right) - \\
& -E_0 \sum \beta'^i \mu_{t+i}' \left(\begin{array}{c} B_{t+i}^{'} - (1-\Upsilon)B_{t+i-1}^{'} - LTV_{t+i}q_{t+i}h_{o,t+i}' \\ -(k-1)\, LTV_{t+i}q_{t+i}h_{o,t+i-1}' + k(1-\Upsilon)B_{t+i-1}^{'} \end{array} \right) \\
& -E_0 \sum \beta'^i \Lambda_{t+i}' \left(R_{t+i}^L - \frac{B_{t+i}^{'} - (1-\Upsilon)B_{t+i-1}^{'}}{B_{t+i}^{'}}(R_{t+i} + \Psi) - (1 - \frac{B_{t+i}^{'} - (1-\Upsilon)B_{t+i-1}^{'}}{B_{t+i}^{'}})R_{t+i-1}^L \right)
\end{aligned}$$

It is easy to see that $(1-\Upsilon)B_{t+i-1}^{'}$ in the budget constraint cancels out. So, we are left with:

$$\begin{aligned}
L_t' \quad = \quad & E_0 \sum \beta'^i \left(z_{t+i}'\Gamma_c' \log(C_{t+i}' - \zeta^c C_{t+i-1}') + \frac{j_{t+i}'}{1-\vartheta} \log\left[\gamma' h_{o,t+i}'^{1-\vartheta} + (1-\gamma')h_{r,t+i}'^{1-\vartheta}\right] - \frac{1}{1+\varrho} N_{t+i}'^{1+\varrho} \right) - \\
& -E_0 \sum \beta'^i \lambda_{t+i}^{'} \left(\begin{array}{c} C_{t+i}' + q_{t+i}h_{o,t+i}' + r_{t+i}h_{r,t+i}' + R_{t+i-1}^L B_{t+i-1}' \\ -B_{t+i}^{'} - q_{t+i}h_{o,t+i-1}' - w_{t+i}'N_{t+i}' \end{array} \right) - \\
& -E_0 \sum \beta'^i \mu_{t+i}' \left(B_{t+i}^{'} - LTV_{t+i}q_{t+i}h_{o,t+i}' - (k-1)\, LTV_{t+i}q_{t+i}h_{o,t+i-1}' + (k-1)(1-\Upsilon)B_{t+i-1}^{'} \right) \\
& -E_0 \sum \beta'^i \Lambda_{t+i}' \left(R_{t+i}^L - (R_{t+i} + \Psi) + \frac{(1-\Upsilon)B_{t+i-1}^{'}}{B_{t+i}^{'}}(R_{t+i} + \Psi) - \frac{(1-\Upsilon)B_{t+i-1}^{'}}{B_{t+i}^{'}} R_{t+i-1}^L \right)
\end{aligned}$$

The optimization is made w.r.t.: $C_t', h_{o,t}', h_{r,t}', N_t', B_t^{'}, R_t^L$.

F.O.C of the borrowers

(1) $C_t' : \Gamma_c'(\frac{z_t^{'}}{C_t' - \zeta^c C_{t-1}'} - \beta'\zeta^c \frac{z_{t+1}'}{C_{t+1}' - \zeta^c C_t'}) - \lambda_t' = 0 =>$

$$\lambda_t' = \Gamma_c'(\frac{z_t'}{C_t' - \zeta^c C_{t-1}'} - \beta'\zeta^c \frac{z_{t+1}'}{C_{t+1}' - \zeta^c C_t'})$$

(2)$h_{o,t}' : j_t' \frac{\gamma' h_{o,t}'^{-\vartheta}}{\gamma' h_{o,t}'^{1-\vartheta} + (1-\gamma')h_{r,t}'^{1-\vartheta}} - \lambda_t' q_t + \beta'\lambda_{t+1}' q_{t+1} + \mu_t' LTV_t q_t + \beta'\mu_{t+1}' (k-1)\, LTV_{t+1}q_{t+1} = 0 =>$

$$U_{h_o',t} = \lambda_t' q_t - \beta'\lambda_{t+1}' q_{t+1} - \mu_t' LTV_t q_t + \beta'\mu_{t+1}' (1-k)\, LTV_{t+1}q_{t+1}$$

where $U_{h_o',t} = j_t' \frac{\gamma' h_{o,t}'^{-\vartheta}}{\gamma' h_{o,t}'^{1-\vartheta} + (1-\gamma')h_{r,t}'^{1-\vartheta}}$ is the marginal utility from ownership.

(3) $h_{r,t} : j_t' \frac{(1-\gamma')h_{r,t}'^{-\vartheta}}{\gamma' h_{o,t}'^{1-\vartheta} + (1-\gamma')h_{r,t}'^{1-\vartheta}} - \lambda_t^{'} r_t = 0 >$

$$U_{r',t} = \lambda_t^{'} r_t$$

where $U_{r',t} = j_t' \frac{(1-\gamma')h_{r,t}'^{-\vartheta}}{\gamma' h_{o,t}'^{1-\vartheta} + (1-\gamma')h_{r,t}'^{1-\vartheta}}$is the marginal utility from rent.

(4) $B'_t : \lambda'_t - \beta'\lambda'_{t+1}R^L_t + \Lambda'_t \frac{(1-\Upsilon)B'_{t-1}}{B'^2_t}[R_t + \Psi - R^L_{t-1}] - \beta'\Lambda'_{t+1}\frac{(1-\Upsilon)}{B'_{t+1}}[(R_{t+1} + \Psi) - R^L_t] - \mu'_t - \beta'\mu'_{t+1}(k-1)(1-\Upsilon) = 0 >$

$$\mu'_t = \lambda'_t - \beta'\lambda'_{t+1}R^L_t + \beta'\mu'_{t+1}(1-k)(1-\Upsilon) + \underbrace{\Lambda'_t \frac{(1-\Upsilon)B'_{t-1}}{B'^2_t}[R_t + \Psi - R^L_{t-1}] - \beta'\Lambda'_{t+1}\frac{(1-\Upsilon)}{B'_{t+1}}[R_{t+1} + \Psi - R^L_t]}_{\text{Internalization effect}}$$

(5) $R^L_t : -\beta'\lambda'_{t+1}B'_t - \Lambda'_t + \beta'\Lambda'_{t+1}\frac{(1-\Upsilon)B'_t}{B'_{t+1}} = 0$

$$\beta'\lambda'_{t+1}B'_t + \Lambda'_t = \beta'\Lambda'_{t+1}\frac{(1-\Upsilon)B'_t}{B'_{t+1}}$$

Notice: Under a variable-rate mortgage, ($R^L_t = R_t + \Psi$), or under a fixed-rate mortgage with no internalization of the effect of new borrowing on the effective interest rate applied to the outstanding stock of debt,$R^L_t = \frac{B'^{NEW}_t}{B'_t}(R_t + \Psi) + (1 - \frac{B'^{NEW}_t}{B'_t})R^L_{t-1}$ the first-order condition (5) would be absent, and the effect of internalization in Eq. (4) would disappear.

(6) $N'_t : -\frac{1+\varrho}{1+\varrho}N'^{\varrho}_t - \lambda'_t(-w'_t) = 0 =>$

$$N'^{\varrho}_t = \lambda'_t w'_t$$

2 Firms

There is a continuum of monopolistically competitive firms indexed by $z \in [0,1]$. Each firm produces a differentiated intermediate good using labor $\bar{N}_t(z)$ with a Cobb–Douglas production function (which is linear in labor). We assume that the share of each type of labor in the production function is identical to the share in the population, τ. There is no capital in production.

$$Y_t(z) = A_t\bar{N}_t(z)$$

where $\bar{N}_t(z) = (N_t(z))^{\tau}(N'_t(z))^{1-\tau}$ is a Cobb-Douglas composite of both types of workers. The specification of the production function is the same as in Beningo et al. (2020). The specification of labor input $\bar{N}_t(z)$ is also the same as in Sun and Tsang (2021), but the production function there also includes capital. The technology shock $\log(A_t)$ follows an $AR(1)$ process.

The differentiated-good firms sell their products in a monopolistic competition to a composite firm, producing the aggregate domestic good:

$$Y_t = \left(\int_0^1 Y_t(z)^{\frac{\eta_t-1}{\eta_t}}dz\right)^{\frac{\eta_t}{\eta_t-1}},$$

where η_t is the time-varying elasticity of substitution between the domestically produced differentiated goods, thus serving as a mark-up shock to domestic inflation π_t. The shock $\log(\eta_t)$ follows an $AR(1)$ process. The aggregate good Y_t is then sold in perfect competition at the price $P_t = \left(\int_0^1 P_t(z)^{1-\eta_t} dz\right)^{\frac{1}{1-\eta_t}}$ and is used for private consumption. Minimizing production costs by the composite firm implies the following demand functions for the differentiated goods:

$$Y_t(z) = [\frac{P_t(z)}{P_t}]^{-\eta_t} Y_t$$

We assume nominal price rigidities à la Rotemberg (1982). Each domestic firm z seeks to maximize its expected profits. The discounting factor of profits depends on firms' ownership. The simplest assumption is that firms are owned by the lenders. This assumption is in line with Gurrieri and Jacaviello (2017). The expected real profits:

$$\max E_t \sum_{i=0}^{\infty} sdf_{t,t+i} x_{t+i}(z),$$

where $x_t(z)$ is a real profit of firm z, and $sdf_{t,t+i} = \beta^i \frac{\lambda_{t+i}}{\lambda_t}$, is a (real) stochastic discount factors of the lenders.

$$x_t(z) = \frac{P_t(z)}{P_t} Y_t(z) - w_t N_t(z) - w'_t N'_t(z) - \frac{\chi}{2}\left(\frac{P_t(z)}{P_{t-1}(z)} - (\pi)^{1-\varpi}\left(\frac{P_{t-1}}{P_{t-2}}\right)^{\varpi}\right)^2 Y_t$$

where π is the SS inflation rate, ϖ is the degree of indexation to past inflation, χ is the cost associated with changing the prices, and $w_t = \frac{W_t}{P_t}, w'_t = \frac{W'_t}{P_t}$ are real wages.

The dynamic optimization problem is given by:

$$\max E_t \sum_{i=0}^{\infty} sdf_{t,t+i} \left\{ \begin{array}{c} [\frac{P_{t+i}(z)}{P_{t+i}}]^{1-\eta} Y_{t+i} - w_{t+i} N_{t+i}(z) - w'_{t+i} N'_{t+i}(z) \\ -\frac{\chi}{2}\left(\frac{P_{t+i}(z)}{P_{t+i-1}(z)} - (\pi)^{1-\varpi}\left(\frac{P_{t+i-1}}{P_{t+i-2}}\right)^{\varpi}\right)^2 Y_{t+i} - \\ -\Gamma_{t+i}(z)[(\frac{P_{t+i}(z)}{P_{t+i}})^{-\eta} Y_{t+i} - A_{t+i}(N_{t+i}(z))^{\tau}(N'_{t+i}(z))^{1-\tau}] \end{array} \right\}$$

where $\Gamma_{t+i}(z)$ is a Lagrangian multiplier of the production function constraint.

(1) The first-order conditions w.r.t. price $P_t(z)$:

$$\begin{aligned} & (1-\eta)[\frac{P_t(z)}{P_t}]^{-\eta}\frac{Y_t}{P_t} - \chi\left(\frac{P_t(z)}{P_{t-1}(z)} - (\pi)^{1-\varpi}\left(\frac{P_{t-1}}{P_{t-2}}\right)^{\varpi}\right)\frac{Y_t}{P_{t-1}(z)} \\ & + sdf_{t,t+1}\chi\left(\frac{P_{t+1}(z)}{P_t(z)} - (\pi)^{1-\varpi}\left(\frac{P_t}{P_{t-1}}\right)^{\varpi}\right) Y_{t+1}\frac{P_{t+1}(z)}{(P_t(z))^2} + \eta\Gamma_t(z)(\frac{P_t(z)}{P_t})^{-\eta-1}\frac{Y_t}{P_t} \\ = \quad & 0 \end{aligned}$$

Lets divide by Y_t

$$
\begin{aligned}
& (1-\eta)[\frac{P_t(z)}{P_t}]^{-\eta}\frac{1}{P_t} - \chi\left(\frac{P_t(z)}{P_{t-1}(z)} - (\pi)^{1-\varpi}\left(\frac{P_{t-1}}{P_{t-2}}\right)^{\varpi}\right)\frac{1}{P_{t-1}(z)} \\
& + sdf_{t,t+1}\chi\left(\frac{P_{t+1}(z)}{P_t(z)} - (\pi)^{1-\varpi}\left(\frac{P_t}{P_{t-1}}\right)^{\varpi}\right)\frac{Y_{t+1}}{Y_t}\frac{P_{t+1}(z)}{(P_t(z))^2} + \eta\Gamma_t(z)(\frac{P_t(z)}{P_t})^{-\eta-1}\frac{1}{P_t} \\
= \ & 0
\end{aligned}
$$

Now we multiply by $P_t(z)$

$$
\begin{aligned}
& (1-\eta)[\frac{P_t(z)}{P_t}]^{-\eta}\frac{P_t(z)}{P_t} - \chi\left(\frac{P_t(z)}{P_{t-1}(z)} - (\pi)^{1-\varpi}\left(\frac{P_{t-1}}{P_{t-2}}\right)^{\varpi}\right)\frac{P_t(z)}{P_{t-1}(z)} \\
& + sdf_{t,t+1}\chi\left(\frac{P_{t+1}(z)}{P_t(z)} - (\pi)^{1-\varpi}\left(\frac{P_t}{P_{t-1}}\right)^{\varpi}\right)\frac{Y_{t+1}}{Y_t}\frac{P_{t+1}(z)}{P_t(z)} + \eta\Gamma_t(z)(\frac{P_t(z)}{P_t})^{-\eta-1}\frac{P_t(z)}{P_t} \\
= \ & 0
\end{aligned}
$$

Now lets define $\pi_t(z) = \frac{P_t(z)}{P_{t-1}(z)}, \pi_t = \frac{P_t}{P_{t-1}}$.

$$
\begin{aligned}
& (1-\eta)[\frac{P_t(z)}{P_t}]^{1-\eta} - \chi\left(\pi_t(z) - (\pi)^{1-\varpi}(\pi_{t-1})^{\varpi}\right)\pi_t(z) \\
& + sdf_{t,t+1}\chi\left(\pi_{t+1}(z) - (\pi)^{1-\varpi}(\pi_t)^{\varpi}\right)\frac{Y_{t+1}}{Y_t}\pi_{t+1}(z) + \eta\Gamma_t(z)(\frac{P_t(z)}{P_t})^{-\eta} \\
= \ & 0
\end{aligned}
$$

Since $P_t(z) = P_t$, and $\pi_t(z) = \pi_t$ (all firms are identical w.r.t. price setting)

$$
(1-\eta) - \chi\left(\pi_t - (\pi)^{1-\varpi}(\pi_{t-1})^{\varpi}\right)\pi_t + sdf_{t,t+1}\chi\left(\pi_{t+1} - (\pi)^{1-\varpi}(\pi_t)^{\varpi}\right)\frac{Y_{t+1}}{Y_t}\pi_{t+1} + \eta\Gamma_t(z) = 0
$$

In fact, this is a Phillips curve (now we put back η_t with index t to empathize that this is time-variant and it is a source of markup shock).

$$
\Gamma_t(z) = \frac{\eta_t - 1}{\eta_t} + \frac{\chi}{\eta_t}\left(\pi_t - (\pi)^{1-\varpi}(\pi_{t-1})^{\varpi}\right)\pi_t - \frac{sdf_{t,t+1}\chi}{\eta_t}\left(\pi_{t+1} - (\pi)^{1-\varpi}(\pi_t)^{\varpi}\right)\frac{Y_{t+1}}{Y_t}\pi_{t+1}
$$

where $sdf_{t,t+1} = \beta\lambda_{t+1}/\lambda_t$ is a stochastic discount factor of unconstrained households.

Notice that in the SS this is a marginal cost (inverse of markup, which is $\frac{\eta}{\eta-1}$):

$$
\Gamma = \frac{\eta - 1}{\eta}
$$

(2) First-order conditions w.r.t. demand for labor $N_t(z)$ and $N_t'(z)$:
$N_t(z) : -w_t + \tau\Gamma_t(z)A_t(N_t(z))^{\tau-1}(N_t'(z))^{1-\tau} = 0 => w_t = \tau\Gamma_t(z)A_t(N_t(z))^{\tau-1}(N_t'(z))^{1-\tau} =>$

$$
\Gamma_t(z) = \frac{w_t}{\tau\frac{Y_t(z)}{N_t(z)}}
$$

This equation is a marginal costs (real wage of the lenders divided by their marginal product of labor)

$N_t'(z)$: $-w_t' + (1-\tau)\Gamma_{t+i}(z)A_{t+i}(N_{t+i}(z))^{\tau}(N_{t+i}'(z))^{-\tau} = 0 => w_t' = (1-\tau)\Gamma_t(z)A_t(N_t(z))^{\tau}(N_t'(z))^{-\tau} =>$

$$\Gamma_t(z) = \frac{w_t'}{(1-\tau)\frac{Y_t(z)}{N_t'(z)}}$$

This equation is a marginal costs (real wage of the borrowers divided by their marginal product of labor).

After dividing two last equations by each other we have:

$$\frac{1-\tau}{\tau}\frac{w_t}{w_t'} = \frac{N_t'(z)}{N_t(z)}$$

We rewrite the last equation as:

$$(1-\tau)w_t N_t(z) = \tau w_t' N_t'(z)$$

Now we aggregate over all firms (over measure 1):

$$(1-\tau)w_t\int_0^1 N_t(z)dz = \tau w_t'\int_0^1 N_t'(z)dz$$

Note that $\int_0^1 N_t(z)dz = \int_0^1 N_t(j)dj = N_t$,where index j denotes the index of the lenders.

Thus:

$$(1-\tau)w_t N_t = \tau w_t' N_t'$$

Now we return to the equation above which is the marginal cost of firms, $\Gamma_t(z)$:

$$\Gamma_t(z) = \frac{w_t}{\tau\frac{Y_t(z)}{N_t(z)}}$$

After simplifying

$$\Gamma_t(z) = \frac{w_t}{\tau}\frac{N_t(z)}{Y_t(z)}$$

Since each firm z produces the same amount of output, $Y_t(z) = Y_t$, we get

$$\Gamma_t(z) = \frac{w_t}{Y_t\tau}N_t(z)$$

Now we aggregate over all firms z of measure 1:

$$\Gamma_t = \int_0^1 \Gamma_t(z)dz = \frac{w_t}{Y_t\tau}\int_0^1 N_t(z)dz = \frac{w_t}{Y_t\tau}\int_0^1 N_t(j)dj = \frac{w_t}{Y_t\tau}N_t$$

or

$$\Gamma_t = \frac{w_t}{\tau Y_t} N_t$$

Finally

$$w_t = \Gamma_t \frac{\tau Y_t}{N_t}$$

Using previous identify we have found $(1-\tau)w_t N_t = \tau w_t' N_t'$ we get:

$$w_t' = \frac{1-\tau}{\tau} \frac{w_t N_t}{N_t'}$$

The wage income of two types of households is: $w_t N_t + w_t' N_t' = w_t N_t + \frac{1-\tau}{\tau} w_t N_t = \frac{1}{\tau} w_t N_t = \frac{1}{\tau} \Gamma_t \frac{\tau Y_t}{N_t} N_t = \Gamma_t Y_t$.

Real dividends (since $P_t^D(z) = P_t^D(z), Y_t(z) = Y_t$) :

$$Div_t = Y_t - w_t N_t - w_t' N_t' - \frac{\chi}{2} \left(\pi_t - (\pi)^{1-\varpi} (\pi_{t-1})^{\varpi} \right)^2 Y_t$$

In the SS, $Div = (1-\Gamma)Y = (1 - \frac{\eta-1}{\eta})Y = \frac{1}{\eta} Y$.

3 Commercial Banks (financial intermediates)

Commercial banks act as intermediaries, accepting deposits from lenders and extending loans to borrowers. The banks place these deposits with the central bank, earning the announced interest rate. Their budget constraint is given by :

$$R_{t-1} B_{t-1}^{'} + R_{t-1}^{L} B_{t-1}^{'} = B_t^{'} + R_{t-1} B_{t-1}^{'} + Div_t^b,$$

where the left-hand side represents income from deposits at the central bank and from loans to borrowers, while the right-hand side represents interest payments to lenders and deposits at the central bank.

We define profits as:

$$Div_t^b = R_{t-1} B_{t-1}^{'} - B_t^{'} + (R_{t-1}^{L} - R_{t-1}) B_{t-1}^{'}.$$

Profits of commercial banks are transferred to lenders, under the assumption that lenders are the sole owners of the commercial banks. Combining the profits of the commercial banks with the profits of the CB (see below) we obtain the net profits transferred to the lenders.

$$Div_t^b = B_{t-1}' (R_{t-1}^{L} - R_{t-1})$$

4 Central bank

The central bank sets the one-period nominal interest rate according to the following Taylor rule:

$$r_t^{CB} = \rho r_{t-1}^{CB} + (1-\rho) \left(r^{CB} + \theta_1 (\pi_t^{CPI} - \pi) + \theta_2 \Delta y_t \right) + \epsilon_t^{CB}$$

where variables without time subscript denote values at the non-stochastic steady state, and ϵ_t^{CB} is a monetary policy shock. Define CPI inflation as $\pi_t^{CPI} = (1-\nu)\pi_t + \nu\pi_t^{rent}$, where π_t is a rate of inflation excluding rent and π_t^{rent} is a inflation of rent services. ν is the weight of rent in the CPI.

The implementation of monetary policy in the model is characterized by the following mechanism:

$$B_t^s = R_{t-1}^s B_{t-1}^s + TR_t,$$

where $B_t^s = -B_t = B_t'$ and $R_t^s = R_t = r_t^{CB} - \pi_{t+1}$.

In this framework, the central bank supplies deposits in the amount demanded by commercial banks and pays interest on these deposits exactly at the announced policy rate. Any resulting profits or losses of the central bank are transferred eventually to the lenders.

4.0.1 Deriving aggregate resource constraint

Combining the household and the clearing conditions in all markets we can arrive at the economy's aggregate resource constraint. The budget constraint of the lenders

$$\begin{aligned} & C_t + q_t(h_{o,t} + h_{inv,t}) + r_t h_{r,t} + R_{t-1}B_{t-1} = \\ = \quad & B_t + q_t(h_{o,t-1} + h_{inv,t-1}) + p_t^m h_{inv,t} + w_t N_t + Div_t \end{aligned}$$

The budget constraint of the borrowers

$$C_t' + q_t h_{o,t}' + r_t h_{r,t}' + (R_{t-1}^L - 1 + \Upsilon)B_{t-1}' = B_t'^{NEW} + q_t h_{o,t-1}' + w_t' N_t'$$

Dividends of retailers, consumption production firms and intermediates:

$$Div_t^{retail} = r_t h_{inv,t} - p_t^m h_{inv,t} - \frac{\Theta}{2}\left(\pi_t^{rent} - (\pi)^{1-\phi}\left(\pi_{t-1}^{rent}\right)^{\phi}\right)^2 r_t h_{inv,t}$$

$$Div_t = Y_t - w_t N_t - w_t' N_t' - \frac{\chi}{2}\left(\pi_t - (\pi)^{1-\varpi}\left(\pi_{t-1}\right)^{\varpi}\right)^2 Y_t$$

$$Div_t^b = B_{t-1}'(R_{t-1}^L - R_{t-1})$$

Summing up two first equations

$$\begin{aligned} & C_t + q_t(h_{o,t} + h_{inv,t}) + r_t h_{r,t} + R_{t-1}B_{t-1} + C_t' + q_t h_{o,t}' + r_t h_{r,t}' + (R_{t-1}^L - 1 + \Upsilon)B_{t-1}' = \\ = \quad & B_t + q_t(h_{o,t-1} + h_{inv,t-1}) + p_t^m h_{inv,t} + w_t N_t + Div_t + B_t'^{NEW} + q_t h_{o,t-1}' + w_t' N_t' \end{aligned}$$

Plug in the dividends:

$$\begin{aligned} & C_t + q_t(h_{o,t} + h_{inv,t}) + r_t h_{r,t} + R_{t-1}B_{t-1} + C_t' + q_t h_{o,t}' + r_t h_{r,t}' + (R_{t-1}^L - 1 + \Upsilon)B_{t-1}' = \\ = \quad & B_t + q_t(h_{o,t-1} + h_{inv,t-1}) + p_t^m h_{inv,t} + w_t N_t + r_t h_{inv,t} - p_t^m h_{inv,t} \\ & -\frac{\Theta}{2}\left(\pi_t^{rent} - (\pi)^{1-\phi}\left(\pi_{t-1}^{rent}\right)^{\phi}\right)^2 r_t h_{inv,t} + Y_t - w_t N_t - w_t' N_t' - \frac{\chi}{2}\left(\pi_t - (\pi)^{1-\varpi}\left(\pi_{t-1}\right)^{\varpi}\right)^2 Y_t \\ & + B_t'^{NEW} + q_t h_{o,t-1}' + w_t' N_t' + B_{t-1}'(R_{t-1}^L - R_{t-1}) \end{aligned}$$

After simplifications:

$$
\begin{aligned}
& C_t + C'_t + q_t(h_{o,t} + h_{inv,t}) + r_t h_{r,t} + R_{t-1}B_{t-1} + q_t h'_{o,t} + r_t h'_{r,t} + (R^L_{t-1} - 1 + \Upsilon)B'_{t-1} = \\
= \quad & B_t + q_t(h_{o,t-1} + h_{inv,t-1}) + r_t h_{inv,t} - \frac{\Theta}{2}\left(\pi^{rent}_t - (\pi)^{1-\phi}\left(\pi^{rent}_{t-1}\right)^{\phi}\right)^2 r_t h_{inv,t} + Y_t \\
& -\frac{\chi}{2}\left(\pi_t - (\pi)^{1-\varpi}\left(\pi_{t-1}\right)^{\varpi}\right)^2 Y_t \\
& + B_t^{'NEW} + q_t h'_{o,t-1} + B'_{t-1}(R^L_{t-1} - R_{t-1})
\end{aligned}
$$

Now take into account that $h_{inv,t} = h_{r,t} + h'_{r,t}$,and $h_{inv,t-1} = h_{r,t-1} + h'_{r,t-1}$, and using $h_{r,t} + h'_{r,t} + h_{o,t} + h'_{o,t} = h$ and $h_{r,t-1} + h'_{r,t-1} + h_{o,t-1} + h'_{o,t-1} = h$)

$$
\begin{aligned}
& C_t + C'_t + R_{t-1}B_{t-1} + (R^L_{t-1} - 1 + \Upsilon)B'_{t-1} = \\
= \quad & Y_t + B_t^{'NEW} + B_t - \frac{\Theta}{2}\left(\pi^{rent}_t - (\pi)^{1-\phi}\left(\pi^{rent}_{t-1}\right)^{\phi}\right)^2 r_t h_{inv,t} - \frac{\chi}{2}\left(\pi_t - (\pi)^{1-\varpi}\left(\pi_{t-1}\right)^{\varpi}\right)^2 Y_t \\
& + B'_{t-1}(R^L_{t-1} - R_{t-1})
\end{aligned}
$$

Taking account that $B_t + B_t^{'} = 0$, $B_{t-1} + B_{t-1}^{'} = 0$.

$$
\begin{aligned}
& C_t + C'_t + R_{t-1}B_{t-1} + (R^L_{t-1} - 1 + \Upsilon)B'_{t-1} - B_t^{'NEW} - B'_{t-1}(R^L_{t-1} - R_{t-1}) - B_t = \\
= \quad & Y_t - \frac{\Theta}{2}\left(\pi^{rent}_t - (\pi)^{1-\phi}\left(\pi^{rent}_{t-1}\right)^{\phi}\right)^2 r_t h_{inv,t} - \frac{\chi}{2}\left(\pi_t - (\pi)^{1-\varpi}\left(\pi_{t-1}\right)^{\varpi}\right)^2 Y_t
\end{aligned}
$$

Now we should prove that:

$$
R_{t-1}B_{t-1} + (R^L_{t-1} - 1 + \Upsilon)B'_{t-1} - B_t^{'NEW} - B'_{t-1}(R^L_{t-1} - R_{t-1}) - B_t = 0
$$

After simplifications:

$$
R_{t-1}B_{t-1} + R^L_{t-1}B'_{t-1} - (1 - \Upsilon)B'_{t-1} - B_t^{'NEW} - B'_{t-1}R^L_{t-1} + B'_{t-1}R_{t-1} - B_t = 0
$$

Using $B'_t = (1 - \Upsilon)B'_{t-1} + B_t^{'NEW}$, and $B_t + B_t^{'} = 0$, we can see that all expressions in brackets are equal to zero:

$$
R_{t-1}(B_{t-1} + B'_{t-1}) + (R^L_{t-1}B'_{t-1} - B'_{t-1}R^L_{t-1}) - (B'_t + B_t) = 0
$$

That means that the aggregate constraint of the economy is:

$$
C_t + C'_t = Y_t - \frac{\chi}{2}\left(\pi_t - (\pi)^{1-\varpi}\left(\pi_{t-1}\right)^{\varpi}\right)^2 Y_t - \frac{\Theta}{2}\left(\pi^{rent}_t - (\pi)^{1-\phi}\left(\pi^{rent}_{t-1}\right)^{\phi}\right)^2 r_t h_{inv,t}
$$

That means that the total resources are equal to total product net of adjustment costs

5 Exogenous processes

The two exogenous processes are modeled as:

$$\log z'_t = \rho_z \log z'_t + u_{z,t} + u'_{z,t}$$

$$\log z_t = \rho_z \log z_t + u_{z,t}$$

There is a common innovation $u_{z,t}$ as well as possible idiosyncractic for constrained group $u'_{z,t}$.

$$\log j'_t = (1 - \rho'_j)\log(j') + \rho'_j \log j'_t + u_{j,t} + u'_{j,t}$$

$$\log j_t = (1 - \rho_j)\log(j) + \rho_j \log j_t + u_{j,t}$$

For simplicity we assume that $j' = j$,and $\rho'_j = \rho_j$.

There is a common innovation $u_{j,t}$ as well as possible idiosyncractic innovation for constrained group $u'_{j,t}$.

The $\log(A_t)$ is a (stationary) technological shock given by:

$$\log(A_t) = (1 - \rho^A)\log(A) + \rho^A \log(A_{t-1}) + \epsilon^a_t$$

There is also markup shock

$$\log(\mu_t) = (1 - \rho^\mu)\log(\mu) + \rho^\mu \log(\mu_{t-1}) + \epsilon^\mu_t$$

The monetary policy shock

$$u^i_t = \rho^i u^i_{t-1} + \epsilon^{u_i}_t$$

6 Summary of FOC and equilibrium

6.1 Lenders

$$\lambda_t = \Gamma_c\left(\frac{z_t}{C_t - \zeta^c C_{t-1}} - \beta\zeta^c \frac{z_{t+1}}{C_{t+1} - \zeta^c C_t}\right) \tag{1}$$

$$j_t \frac{\gamma h_{o,t}^{-\vartheta}}{\gamma h_{o,t}^{1-\vartheta} + (1-\gamma)h_{r,t}^{1-\vartheta}} = \lambda_t q_t - \beta\lambda_{t+1} q_{t+1} \tag{2}$$

$$j_t \frac{(1-\gamma) h_{r,t}^{-\vartheta}}{\gamma h_{o,t}^{1-\vartheta} + (1-\gamma)h_{r,t}^{1-\vartheta}} = \lambda_t r_t \tag{3}$$

$$\lambda_t = \beta\lambda_{t+1} R_t \tag{4}$$

$$N_t^{\varrho} = \lambda_t w_t \tag{5}$$

$$p_t^m = q_t - sdf_{t,t+1} q_{t+1} \tag{6}$$

$$C_t + q_t(h_{o,t} + h_{inv,t}) + r_t h_{r,t} + R_{t-1} B_{t-1} = B_t + q_t(h_{o,t-1} + h_{inv,t-1}) + p_t^m h_{inv,t} + w_t N_t + Div_t \tag{7}$$

6.2 Borrowers

$$\lambda_t' = \Gamma_c'\left(\frac{z_t'}{C_t' - \zeta^c C_{t-1}'} - \beta' \zeta^c \frac{z_{t+1}'}{C_{t+1}' - \zeta^c C_t'}\right) \tag{8}$$

$$j_t' \frac{\gamma' h_{o,t}'^{-\vartheta}}{\gamma' h_{o,t}'^{1-\vartheta} + (1-\gamma') h_{r,t}'^{1-\vartheta}} = \lambda_t' q_t - \beta' \lambda_{t+1}' q_{t+1} - \mu_t' LTV_t q_t + \beta' \mu_{t+1}' (1-k) LTV_{t+1} q_{t+1} \tag{9}$$

$$j_t' \frac{(1-\gamma') h_{r,t}'^{-\vartheta}}{\gamma' h_{o,t}'^{1-\vartheta} + (1-\gamma') h_{r,t}'^{1-\vartheta}} = \lambda_t^{'} r_t \tag{10}$$

Under fixed-rate mortgage and internalization:

$$\begin{aligned} \mu_t' \quad = \quad & \lambda_t^{'} - \beta' \lambda_{t+1}^{'} R_t^L + \beta' \mu_{t+1}' (1-k)(1-\Upsilon) \\ & \underbrace{+ \Lambda_t' \frac{(1-\Upsilon) B_{t-1}^{'}}{B_t^{'2}} [R_t + \Psi - R_{t-1}^L] - \beta' \Lambda_{t+1}' \frac{(1-\Upsilon)}{B_{t+1}^{'}} [R_{t+1} + \Psi - R_t^L]}_{\text{Internalization effect}} \end{aligned} \tag{11}$$

$$\beta' \lambda_{t+1}^{'} B_t' + \Lambda_t' = \beta' \Lambda_{t+1}' \frac{(1-\Upsilon) B_t^{'}}{B_{t+1}^{'}} \tag{12}$$

Under variable-rate mortgage or under fixed-rate mortgage with no internalization (Eq. (12) would be absent):

$$\mu_t' = \lambda_t^{'} - \beta' \lambda_{t+1}^{'} R_t^L + \beta' \mu_{t+1}' (1-k)(1-\Upsilon) \tag{13}$$

$$N_t'^{\varrho} = \lambda_t^{'} w_t' \tag{14}$$

$$B_t^{'} = (1-\Upsilon) B_{t-1}^{'} + B_t^{'NEW} \tag{15}$$

$$B_t^{'NEW} = LTV_t q_t (h_{o,t}' - h_{o,t-1}') + k\left(LTV_t q_t h_{o,t-1}' - (1-\Upsilon) B_{t-1}^{'}\right) \tag{16}$$

$$C_t' + q_t h_{o,t}' + r_t h_{r,t}' + (R_{t-1}^L - 1 + \Upsilon) B_{t-1}' = B_t^{'NEW} + q_t h_{o,t-1}' + w_t' N_t' \tag{17}$$

6.3 Retailers

$$\frac{p_t^m}{r_t} = \frac{\eta_r - 1}{\eta_r} + \frac{\Theta}{\eta_r}\pi_t^{rent}\left(\pi_t^{rent} - (\pi)^{1-\phi}\left(\pi_{t-1}^{rent}\right)^{\phi}\right) - sdf_{t,t+1}^r\frac{\Theta}{\eta_r}\frac{Y_{t+1}^{rent}}{Y_t^{rent}}\pi_{t+1}^{rent}\left(\pi_{t+1}^{rent} - (\pi)^{1-\phi}\left(\pi_t^{rent}\right)^{\phi}\right) \tag{18}$$

$$\pi_t^{rent} = \frac{r_t}{r_{t-1}}\pi_t \tag{19}$$

$$Div_t^{retail} = r_t h_{inv,t} - p_t^m h_{inv,t} - \frac{\Theta}{2}\left(\pi_t^{rent} - (\pi)^{1-\phi}\left(\pi_{t-1}^{rent}\right)^{\phi}\right)^2 r_t h_{inv,t} \tag{20}$$

6.4 Firms

$$\Gamma_t = \frac{\eta_t - 1}{\eta_t} + \frac{\chi}{\eta_t}\pi_t\left(\pi_t - (\pi)^{1-\varpi}(\pi_{t-1})^{\varpi}\right) - sdf_{t,t+1}\frac{\chi}{\eta_t}\frac{Y_{t+1}}{Y_t}\pi_{t+1}\left(\pi_{t+1} - (\pi)^{1-\varpi}(\pi_t)^{\varpi}\right) \tag{21}$$

$$Y_t = A_t(N_t)^{\tau}(N_t')^{1-\tau} \tag{22}$$

$$Div_t = Y_t - w_t N_t - w_t' N_t' - \frac{\chi}{2}\left(\pi_t - (\pi)^{1-\varpi}(\pi_{t-1})^{\varpi}\right)^2 Y_t \tag{23}$$

$$w_t = \Gamma_t \frac{\tau Y_t}{N_t} \tag{24}$$

$$w_t' = \frac{1-\tau}{\tau}\frac{w_t N_t}{N_t'} \tag{25}$$

6.5 Commercial banks

$$R_t^L = R_t + \Psi \tag{26}$$

or alternatively

$$R_t^L = \frac{B_t^{'NEW}}{B_t^{'}}(R_t + \Psi) + (1 - \frac{B_t^{'NEW}}{B_t^{'}})R_{t-1}^L$$

$$Div_t^b = (R_{t-1}^L - R_{t-1})B_{t-1}' \tag{27}$$

6.6 Central bank

$$r_t^{CB} = \rho r_{t-1}^{CB} + (1-\rho)\left(r^{CB} + \theta_1(\pi_t^{CPI} - \pi) + \theta_2 \Delta y_t\right) + \epsilon_t^{CB} \tag{28}$$

$$\pi_t^{CPI} = (1-\nu)\pi_t + \nu\pi_t^{rent} \tag{29}$$

$$R_t = r_t^{CB} - \pi_{t+1} \tag{30}$$

$$LTV_t = (1-\rho^{LTV})LTV + \rho^{LTV}LTV_{t-1} + \epsilon_t^{LTV} \tag{31}$$

6.7 Market clearing

$$C_t+C_t^{\prime} = Y_t-\frac{\chi}{2}\left(\pi_t - (\pi)^{1-\varpi}\left(\pi_{t-1}\right)^{\varpi}\right)^2 Y_t-\frac{\Theta}{2}\left(\pi_t^{rent} - (\pi)^{1-\phi}\left(\pi_{t-1}^{rent}\right)^{\phi}\right)^2 r_t h_{inv,t} \tag{32}$$

$$B_t + B_t^{'} = 0 \tag{33}$$

$$h_{inv,t} = h_{r,t} + h_{r,t}^{\prime} \tag{34}$$

$$h_{r,t} + h_{r,t}^{\prime} + h_{o,t} + h_{o,t}^{\prime} = h \tag{35}$$

****

7 Deriving SS

From the FOC of the lenders: $\lambda_t = \frac{1-\zeta^c}{1-\beta\zeta^c}(\frac{z_t}{C_t - \zeta^c C_{t-1}} - \beta\zeta^c \frac{z_{t+1}}{C_{t+1} - \zeta^c C_t})$

$$\lambda = \frac{1}{C}$$

The same holds for the borrowers:

$$\lambda' = \frac{1}{C'}$$

From Equation (5) in the first order condition of the borrowers:

$$\beta' \lambda^{'} B' + \Lambda' = \beta' \Lambda' \frac{(1-\Upsilon)B^{'}}{B'}$$

Thus

$$\Lambda' = \frac{\lambda^{'} B'}{1 - \Upsilon - \frac{1}{\beta'}}$$

From FOC of the lenders $\lambda_t = \beta \lambda_{t+1} R$, we get in SS:

$$R = \frac{1}{\beta}$$

From the FOC of the firms $\Gamma_t = \frac{\eta_t - 1}{\eta_t} + \frac{\chi}{\eta_t}\left(\pi_t - (\pi)^{1-\varpi}(\pi_{t-1})^{\varpi}\right)\pi_t - \frac{\beta\chi}{\eta_t}\left(\pi_{t+1} - (\pi)^{1-\varpi}(\pi_t)^{\varpi}\right)\frac{Y_{t+1}}{Y_t}\pi_{t+1}$ we get in SS the real marginal cost (inverse of markup):

$$\Gamma = \frac{\eta - 1}{\eta}$$

From demand of the firms for labor we get

$$\Gamma = \frac{w'}{(1-\tau)\frac{Y}{N'}}$$

For the borrowers we obtain

$$w'N' = (1-\tau)\Gamma Y$$

We also obtain for the lenders

$$wN = \tau \Gamma Y$$

From the supply of labor of the borrowers we obtain

$$N'^{\varrho} = \frac{1}{C'} w'$$

From the budget constraint of the borrowers

$$C' + rh'_r + (R^L - 1 + \Upsilon)B' = B'^{NEW} + w'N'$$

where $R^L = R + \Psi$.
Using the definition of debt

$$B' = (1-\Upsilon)B' + B'^{NEW}$$

From here the new debt:

$$B'^{NEW} = \Upsilon B'$$

Combining with the supply of new debt we obtain

$$B^{'} = \left\{ \frac{k}{\Upsilon + k(1-\Upsilon)} \right\} LTVqh'_o$$

Using previous formula for $B'^{NEW} = \Upsilon B'$

$$B^{'NEW} = \Upsilon B' = \left\{ \frac{\Upsilon k}{\Upsilon + k(1-\Upsilon)} \right\} LTVqh'_o$$

From the demand of the borrowers for ownership

$$j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}} = \lambda' q - \beta' \lambda' q - \mu' LTVq + \beta' \mu' \left(1-k\right) LTVq$$

Or

$$j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}} = (1-\beta')\lambda' q + LTVq\mu'(\beta' \left(1-k\right) - 1)$$

Finally

$$j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}} = (1-\beta')\lambda' q + (\beta' - \beta' k - 1)\mu' LTVq$$

where $j' = j$ in the SS.
From the demand of the borrowers for debt

$$\mu' = \lambda^{'} - \beta' \lambda^{'} R^L + \beta' \mu' (1-k)(1-\Upsilon)$$

Or

$$\mu' \left[1 - \beta'(1-k)(1-\Upsilon)\right] = \lambda^{'}(1 - \beta' R^L)$$

After simplification

$$\mu' = \frac{\lambda^{'}(1-\beta' R^L)}{1 - \beta'(1-k)(1-\Upsilon)}$$

For further purposes we obtain the ratio

$$\frac{\mu'}{\lambda'} = \frac{1-\beta' R^L}{1-\beta'(1-k)(1-\Upsilon)}$$

Plugging μ' into the demand for ownership of the borrowers $j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}} = (1-\beta')\lambda' q + (\beta' - \beta' k - 1)\mu' LTVq$

$$j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}} = (1-\beta')\lambda' q + (\beta' - \beta' k - 1)LTVq \frac{\lambda^{'}(1-\beta' R^L)}{1-\beta'(1-k)(1-\Upsilon)}$$

Using $\lambda' = \frac{1}{C'}$

$$j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}} = (1-\beta')\frac{q}{C'} + (\beta' - \beta' k - 1)LTV \frac{q}{C'} \frac{1-\beta' R^L}{1-\beta'(1-k)(1-\Upsilon)}$$

Isolating $\frac{q}{C'}$

$$j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}} = \frac{q}{C'}\left[1-\beta' + (\beta' - \beta' k - 1)LTV \frac{\mu'}{\lambda'}\right]$$

Thus we obtain

$$\frac{q}{C'} = \frac{j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}}}{1-\beta' + (\beta' - \beta' k - 1)LTV \frac{\mu'}{\lambda'}}$$

For simplicity of notation let us define, $\theta = 1-\beta' + (\beta' - \beta' k - 1)LTV \frac{\mu'}{\lambda'}$. Plug in the previous expression

$$\frac{q}{C'} = \frac{j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}}}{\theta}$$

For further purposes we multiply both sides by h_0'

$$\frac{qh_0'}{C'} = \frac{j' \frac{\gamma' h_o'^{1-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma')h_r'^{1-\vartheta}}}{\theta}$$

Dividing the expression in the nominator by $\gamma' h_o'^{1-\vartheta}$

$$\frac{qh_0'}{C'} = \frac{j' \frac{1}{1+\frac{1-\gamma'}{\gamma'}\left(\frac{h_r'}{h_o'}\right)^{1-\vartheta}}}{\theta}$$

Finally we obtain

$$\frac{qh_0'}{C'} = \varkappa = \frac{1}{\theta} \frac{j'}{1+\frac{1-\gamma'}{\gamma'}\left(\frac{h_0'}{h_r'}\right)^{\vartheta-1}}$$

Now we derive the ownership-rent ratio for borrowers $\frac{h_0'}{h_r'}$.

From demand for ownership and rent we divide by each other:

$$\frac{j' \frac{\gamma' h_o'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma') h_r'^{1-\vartheta}}}{j' \frac{(1-\gamma') h_r'^{-\vartheta}}{\gamma' h_o'^{1-\vartheta} + (1-\gamma') h_r'^{1-\vartheta}}} = \frac{\lambda' q - \beta' \lambda' q - \mu' LTVq + \beta' \mu' (1-k) LTVq}{\lambda' r}$$

After some algebra

$$\frac{\gamma'}{1-\gamma'} \left(\frac{h_o'}{h_r'}\right)^{-\vartheta} = \frac{\lambda' q - \beta' \lambda' q - \mu' LTVq + \beta' \mu' (1-k) LTVq}{\lambda' r}$$

After dividing by $\lambda' r$ we obtain

$$\frac{\gamma'}{1-\gamma'} \left(\frac{h_o'}{h_r'}\right)^{-\vartheta} = \theta \frac{q}{r}$$

After simplification:

$$\left(\frac{h_o'}{h_r'}\right)^{-\vartheta} = \frac{1-\gamma'}{\gamma'} \frac{q}{r} \theta$$

Finally

$$\Phi^B = \frac{h_o'}{h_r'} = \left[\frac{1-\gamma'}{\gamma'} \frac{q}{r} \theta\right]^{-\frac{1}{\vartheta}}$$

where $\theta = 1 - \beta' + (\beta' - \beta' k - 1) LTV \frac{\mu'}{\lambda'}$(see above) and $\frac{q}{r} = \frac{\frac{\eta_r - 1}{\eta_r}}{1-\beta}$ (see proof below).

Therefore now one can obtain $\frac{qh_0'}{C'}$ using $\frac{h_r'}{h_o'}$ that we have showed previously

$$\frac{qh_0'}{C'} = \varkappa = \frac{1}{\theta} \frac{j'}{1 + \frac{1-\gamma'}{\gamma'} \left(\Phi^B\right)^{\vartheta - 1}}$$

where $\Phi^B = \frac{h_0'}{h_r'}$ is a parameter shown previously.

From here it is easy to express: $qh_0' = \varkappa C'$, where $\varkappa$ is defined previously.

Now we focus on the budget constraint of the borrowers and exploit $(1-\tau)\Gamma Y = w'N'$

$$C' + rh_r' + (R^L - 1 + \Upsilon) B' = B'^{NEW} + (1-\tau)\Gamma Y$$

where $R^L = R + \Psi$.

Now we have

$$C' + rh_r' + (R^L - 1 + \Upsilon) B' = \Upsilon B' + (1-\tau)\Gamma Y$$

or

$$C' + rh_r' + (R^L - 1 + \Upsilon) B' - \Upsilon B' = (1-\tau)\Gamma Y$$

or

$$C' + rh'_r + (R^L - 1)B' = (1-\tau)\Gamma Y$$

Now we use $B^{'} = \delta LTV qh'_o$,where $\delta = \frac{k}{\Upsilon + k(1-\Upsilon)}$ (defined previously).

After plug in B'

$$C' + rh'_r + (R^L - 1)\delta LTV qh'_o = (1-\tau)\Gamma Y$$

Using $qh'_o = \varkappa C'$ (see expression above)

$$C' + rh'_r + \delta(R^L - 1)LTV\varkappa C' = (1-\tau)\Gamma Y$$

After simplifications

$$C'\left[1 + \delta(R^L - 1)LTV\varkappa\right] + rh'_r = (1-\tau)\Gamma Y$$

To complete the budget constraint we need to derive rh'_r.

First note that we have derived above: $\frac{h'_o}{h'_r} = \Phi^B$ and $\frac{qh'_0}{C'} = \varkappa$. After dividing two expressions by each other we get $\frac{\frac{qh'_0}{C'}}{\frac{h'_o}{h'_r}} = \frac{\varkappa}{\Phi^B} => \frac{qh'_r}{C'}$.

Or

$$qh'_r = \frac{\varkappa}{\Phi^B}C'$$

Now we multiply each side by $\frac{r}{q}$, where $\frac{r}{q} = \frac{1-\beta}{\frac{\eta_r - 1}{\eta_r}}$.

$$\frac{r}{q}qh'_r = \frac{\varkappa}{\Phi^B}C'\frac{r}{q}$$

Finally

$$rh'_r = \frac{\varkappa}{\Phi^B}\frac{r}{q}C'$$

Now we plug in rh'_r into the budget constraint of the borrowers

$$C'\left[1 + \delta(R^L - 1)LTV\varkappa\right] + \frac{\varkappa}{\Phi^B}\frac{r}{q}C' = (1-\tau)\Gamma Y$$

After simplifications:

$$C'\left[1 + \delta(R^L - 1)LTV\varkappa + \frac{\varkappa}{\Phi^B}\frac{r}{q}\right] = (1-\tau)\Gamma Y$$

Finally

$$\frac{C'}{Y} = F = \frac{(1-\tau)\Gamma}{1 + \delta(R^L - 1)LTV\varkappa + \Omega}$$

where $\Omega = \frac{\varkappa}{\Phi^B}\frac{r}{q}$.

From labor supply of the borrowers: $N'^{1+\varrho} = \frac{1}{C'} w' N' = \frac{(1-\tau)\Gamma Y}{C'}$

$$C' = \frac{(1-\tau)\Gamma Y}{N'^{1+\varrho}}$$

Or

$$\digamma = \frac{C'}{Y} = \frac{(1-\tau)\Gamma}{N'^{1+\varrho}}$$

Or

$$N'^{1+\varrho} = \frac{(1-\tau)\Gamma}{\digamma}$$

Then

$$N' = \left[\frac{(1-\tau)\Gamma}{\digamma}\right]^{\frac{1}{1+\varrho}}$$

From the resource constraint of the economy

$$C' + C = Y$$

Using $N^{1+\varrho} = \lambda w N => \frac{1}{C}\tau\Gamma Y =>$

$$C = \frac{\tau\Gamma Y}{N^{1+\varrho}} \text{ and } C' = \frac{(1-\tau)\Gamma Y}{N'^{1+\varrho}}$$

Plugging into the resource constraint of the economy $\frac{\tau\Gamma Y}{N^{1+\varrho}} + \frac{(1-\tau)\Gamma Y}{N'^{1+\varrho}} = Y =>$ $\frac{\tau}{N^{1+\varrho}} + \frac{1-\tau}{N'^{1+\varrho}} = \frac{1}{\Gamma}$ we obtain

$$N = \left[\frac{\frac{1}{\Gamma} - \frac{1-\tau}{N'^{1+\varrho}}}{\tau}\right]^{-\frac{1}{1+\varrho}}$$

where N' was found before.
Since N' and N are defined we can find the output

$$Y = (N)^{\tau}(N')^{1-\tau}$$

Then using $\frac{C'}{Y} = \digamma$ we find the consumption of the borrowers

$$C' = \digamma Y$$

Using C' we find the stock of debt (using $B^{'} = \delta LTV q h_o'$ and $\frac{q h_0'}{C'} = \varkappa$)

$$B^{'} = \delta LTV \varkappa C'$$

The consumption of lenders is given by

$$C = Y - C'$$

Now we find wages

$$w = \frac{\tau\Gamma Y}{N}$$

$$w' = \frac{(1-\tau)\Gamma Y}{N'}$$

From demand for housing ownership of lenders

$$j\frac{\gamma h_o^{-\vartheta}}{\gamma h_o^{1-\vartheta} + (1-\gamma)h_r^{1-\vartheta}} = \frac{q}{C}(1-\beta)$$

Or

$$\frac{q}{C} = \frac{j\frac{\gamma h_o^{-\vartheta}}{\gamma h_o^{1-\vartheta}+(1-\gamma)h_r^{1-\vartheta}}}{1-\beta}$$

Let us multiply by h_0 and then divide the numerator and denominator of the RHS by $\gamma h_o^{1-\vartheta}$

$$\frac{qh_0}{C} = \frac{1}{1-\beta}\frac{j}{1+\frac{1-\gamma}{\gamma}\left(\frac{h_0}{h_r}\right)^{\vartheta-1}}$$

From demand for rent of the lenders

$$\frac{r}{C} = j\frac{(1-\gamma)h_r^{-\vartheta}}{\gamma h_o^{1-\vartheta} + (1-\gamma)h_r^{1-\vartheta}}$$

Dividing equation for $\frac{r}{C}$ by $\frac{q}{C}$ (above), $\frac{r}{q} = \frac{\frac{r}{C}}{\frac{q}{C}}$ we define $\frac{r}{q}$:

$$\frac{\frac{r}{C}}{\frac{q}{C}} = (1-\beta)\frac{(1-\gamma)h_r^{-\vartheta}}{\gamma h_o^{-\vartheta}}$$

Then

$$\frac{r}{q} = \frac{1-\gamma}{\gamma}(1-\beta)\left(\frac{h_o}{h_r}\right)^{\vartheta}$$

From investor FOC we have $\lambda(q-p^m) = \beta\lambda q$, thus:

$$\frac{p^m}{q} = 1-\beta$$

where

$$p^m = \left(\frac{\eta_r - 1}{\eta_r}\right) r$$

Combining two previous equations we get

$$\frac{r}{q} = \frac{1-\beta}{\frac{\eta_r-1}{\eta_r}}$$

Now we return to the ratios $\frac{r}{q}$ from two types of households ($\frac{r}{q} = \frac{1-\gamma}{\gamma}(1-\beta)\left(\frac{h_o}{h_r}\right)^{\vartheta}$.

First we derive for the lenders

$$\frac{r}{q} = \frac{1-\beta}{\frac{\eta_r - 1}{\eta_r}} = \frac{1-\gamma}{\gamma}(1-\beta)\left(\frac{h_o}{h_r}\right)^{\vartheta}$$

Then

$$\frac{\eta_r}{\eta_r - 1} = \frac{1-\gamma}{\gamma}\left(\frac{h_o}{h_r}\right)^{\vartheta}$$

The ownership-rent ratio of the lenders

$$\Phi^L = \frac{h_o}{h_r} = \left[\frac{\eta_r}{\eta_r - 1}\frac{\gamma}{1-\gamma}\right]^{\frac{1}{\vartheta}}$$

Thus we can derive

$$\frac{qh_0}{C} = \frac{1}{1-\beta}\frac{j}{1+\frac{1-\gamma}{\gamma}\left(\Phi^L\right)^{\vartheta-1}}$$

For the lenders, using the ratio $\frac{qh_0}{C} = \frac{1}{1-\beta}\frac{j}{1+\frac{1-\gamma}{\gamma}\left(\Phi^L\right)^{\vartheta-1}}$, we derive qh_0

$$s^L = \frac{qh_0}{C} = \frac{1}{1-\beta}\frac{j}{1+\frac{1-\gamma}{\gamma}\left(\Phi^L\right)^{\vartheta-1}}$$

Now we derive $\frac{qh_r}{C}$.
We use equation for $s^L = \frac{qh_0}{C}$ and $\Phi^L = \frac{h_o}{h_r}$ above, noting that $\frac{qh_r}{C} = \frac{s^L}{\Phi^L}$).

$$\frac{qh_r}{C} = \frac{s^L}{\Phi^L}$$

Now we find amount of houses for ownership and rent for all

$$h_r + h_r' + h_o + h_o' = h$$

We have derived previously: $\Phi^L = \frac{h_o}{h_r}, \Phi^B = \frac{h_o'}{h_r'}$.
We also derived: $\varrho^L = \frac{s^L}{\Phi^L} = \frac{qh_r}{C}$,and $\varrho^B = \frac{\varkappa}{\Phi^B} = \frac{qh_r'}{C'}$.
Now we divide two last terms: $\frac{\varrho^L}{\varrho^B} = \frac{\frac{qh_r}{C}}{\frac{qh_r'}{C'}} = \frac{h_r}{h_r'}\frac{C'}{C}$ =>

$$\chi^{LB} = \frac{h_r}{h_r'} = \frac{\varrho^L}{\varrho^B}\frac{C}{C'}$$

where $\varrho^L, \varrho^B, C, C'$ are defined previously.
Thus

$$h_r = \chi^{LB}h_r'$$

Now we derive $\Phi^B = \frac{h_o'}{h_r'}$ => $h_o' = \Phi^B h_r'$, and $\Phi^L = \frac{h_o}{h_r}$ => $h_o = \Phi^L h_r$,where $h_r = \chi^{LB}h_r'$. Thus, $h_o = \Phi^L\chi^{LB}h_r'$.

Plug into $h_r + h'_r + h_o + h'_o = h$

$$\chi^{LB} h'_r + h'_r + \Phi^L \chi^{LB} h'_r + \Phi^B h'_r = h$$

Finally we obtain the amount of houses for rent and ownership

$$h'_r = \frac{h}{1 + \chi^{LB} + \Phi^L \chi^{LB} + \Phi^B}$$

$$h_r = \chi^{LB} \frac{h}{1 + \chi^{LB} + \Phi^L \chi^{LB} + \Phi^B}$$

$$h'_o = \Phi^B \frac{h}{1 + \chi^{LB} + \Phi^L \chi^{LB} + \Phi^B}$$

$$h_o = \Phi^L \chi^{LB} \frac{h}{1 + \chi^{LB} + \Phi^L \chi^{LB} + \Phi^B}$$

Now we calculate the price q

$$q = s^L \frac{C}{h_0}$$

and the price of rent

$$r = \frac{1-\beta}{\frac{\eta_r - 1}{\eta_r}} q$$

and also the price p^m

$$p^m = q(1-\beta)$$

Now we want to show how does the share of the borrowers $1-\tau$ affect their amount of housing consumed.

Let us calculate the overall housing of the borrowers

$$h'_r + h'_o = \frac{1 + \Phi^B}{1 + (1 + \Phi^L)\chi^{LB} + \Phi^B} h$$

The question is how does the ratio $\frac{1+\Phi^B}{1+(1+\Phi^L)\chi^{LB}+\Phi^B}$ depend on the share of the borrowers $1-\tau$?

Note that $\Phi^B = \frac{h'_o}{h'_r}$ and $\Phi^L = \frac{h_o}{h_r}$ are independent on the parameter τ.

Now we examine the parameter χ^{LB}.

$$\chi^{LB} = \frac{\varrho^L}{\varrho^B} \frac{C}{C'}$$

where $\varrho^L = \frac{s^L}{\Phi^L} = \frac{q h_r}{C}$,and $\varrho^B = \frac{\varkappa}{\Phi^B} = \frac{q h'_r}{C'}$.

The parameter $\varrho^L = \frac{s^L}{\Phi^L}$ is independent on τ (since s^L and Φ^L independent on τ).

The parameter $\varrho^B = \frac{\varkappa}{\Phi^B}$is also independent on τ, thus $\frac{\varrho^L}{\varrho^B}$ is independent on τ.

But the ratio$\frac{C}{C'}$ does depend on τ (through the share in the production function) making χ^{LB} be dependent on τ.

To show this we rewrite two consumptions derived previously

$$C = \frac{\tau\Gamma Y}{N^{1+\varrho}} \text{ and } C' = \frac{(1-\tau)\Gamma Y}{N'^{1+\varrho}}$$

Thus

$$\frac{C}{C'} = \frac{\frac{\tau\Gamma Y}{N^{1+\varrho}}}{\frac{(1-\tau)\Gamma Y}{N'^{1+\varrho}}} = \frac{\tau}{1-\tau}\frac{N'^{1+\varrho}}{N^{1+\varrho}}$$

Using

$$N'^{1+\varrho} = \frac{(1-\tau)\Gamma}{\digamma}$$

$$\frac{C}{C'} = \frac{\tau}{1-\tau}\frac{\frac{(1-\tau)\Gamma}{\digamma}}{N^{1+\varrho}} = \tau\frac{\Gamma}{\digamma}\frac{1}{N^{1+\varrho}}$$

Now we use

$$N^{1+\varrho} = \left[\frac{\frac{1}{\Gamma} - \frac{1-\tau}{N'^{1+\varrho}}}{\tau}\right]^{-\frac{1}{1+\varrho}(1+\varrho)} => \frac{\tau\Gamma}{1-\digamma}$$

Thus

$$\frac{C}{C'} = \frac{1-\digamma}{\digamma} = \frac{1}{\digamma} - 1$$

where $\digamma = (1-\tau)\frac{\Gamma}{1+(R^L-1)\left\{\frac{k}{\Upsilon+k(1-\Upsilon)}\right\}LTV\varkappa+\frac{\varkappa}{\Phi^B}\frac{r}{q}}$ depends on τ.

In summary, as the share of borrowers in the population increases, so do their income and share of total output, leading to higher consumption of housing—both owned and rented.